Quadruped SLAM and exploration

Measured results and comparison with published systems | 1 run(s) | simulation

Test conditions

Item	Value
Platform	CoppeliaSim simulation, multi-room arena (10 x 10 m outer walls, 15 inner wall segments, 13 objects)
Robot and sensing	quadruped; forward RGB-D camera converted to a 2D scan; gait odometry; simulated gyro
Algorithm	2D scan-matching SLAM with loop closure and a 2D occupancy-grid map; CPU only
Ground truth	simulator pose of the robot body; wall and object geometry read from the scene (scoring only)
Computer	AMD Ryzen 7 7735HS with Radeon Graphics
Run 1	20261024_210638: safety limit 1800 s, ran 1800 s, timeout (1800s) before all frontiers were cleared

1. Task and exploration

Metric	Run 1	What it measures
Task completed	no	Whether the mission reported every reachable frontier explored
How the run ended	timeout (1800s) before all frontiers were cleared	Stop condition that ended the run
Simulated time	1800 s	Robot time of the run
Distance travelled	157.9 m	Path length of the robot from ground truth
Average speed	0.088 m/s	Distance travelled divided by simulated time
Rooms completed / found	6 / 8	Rooms marked fully explored / rooms the mapper segmented
Real floor mapped as free	88.2 %	Share of the real floor area of the arena marked as free space in the map
Frontiers remaining	48	Boundaries between known and unknown space left at the end
Stuck events	4	Times the anti-stuck watchdog fired
Unreachable targets	9	Targets the path planner could not reach

2. SLAM performance summary (localization and mapping together)

Metric	Run 1	What it measures
Pose error, ATE RMSE	31.0 cm	Root-mean-square position error over the run
Local pose error, RPE	7.4 % / 1.30 deg/m	Translation and heading error of the motion over each 1 m of path
Map accuracy vs real scene	10.6 cm	Mean distance from mapped obstacle cells to the nearest real wall or object
Map completion vs real scene	9.2 cm	Mean distance from real wall and object cells to the nearest mapped obstacle cell
Map F1 vs real scene	77.7 %	Agreement of mapped and real obstacle cells within 15 cm
SLAM update time (median)	10.6 ms	Time of one pose update
Map update time (median)	20.3 ms	Time to integrate one depth frame into the map
Map memory	0.19 MB	Size of the occupancy grid

3. Localization (against ground truth)

Metric	Run 1	What it measures
ATE RMSE	31.0 cm	Root-mean-square position error over every control step (0.05 s), no alignment
ATE median / maximum	27.1 / 63.1 cm	Median and worst position error
ATE as share of path	0.20 %	ATE RMSE divided by distance travelled
Final error	38.5 cm	Position error at the end of the run
RPE translation	9.3 cm per 1 m	Error of the motion over each 1 m of true path
RPE rotation	1.30 deg per m	Heading error of the motion over each 1 m of true path
Heading error mean / maximum	0.74 / 4.9 deg	Orientation error against ground truth
Loop closures	68	Revisits recognized and used to correct drift
Scan-match acceptance	26.9 %	Share of SLAM updates in which the scan match passed its checks

4. Mapping (against the real scene and against a map built with the true pose)

Metric	Run 1	What it measures
Accuracy vs real scene	10.6 cm	Mapped obstacle cells to the nearest real wall or object (mean)
Completion vs real scene	9.2 cm	Real wall and object cells to the nearest mapped obstacle cell (mean)
Precision / recall vs real scene	73.9 / 81.8 %	Mapped cells near a real obstacle / real obstacle cells found, within 15 cm
Free space mapped outside the arena	0.59 m2	Area marked free beyond the outer wall
Accuracy vs true-pose map	12.9 cm	Same as above, against a map built from the same sensor with the true pose
Completion ratio vs true-pose map	54.1 %	Share of true-pose map wall cells matched within 5 cm
F1 vs true-pose map	77.2 %	Wall-cell agreement with the true-pose map within 15 cm
Free-space IoU vs true-pose map	81.2 %	Overlap of mapped free space with true-pose free space

5. Object detection (against the real objects in the scene)

Metric	Run 1	What it measures
Real objects in the scene	13	Objects within the robot's sensing height (0.04-0.60 m)
Detections reported	113	Object entities the robot added to its knowledge graph
Detections on a real object	21.2 %	Share of detections within 0.5 m of a real object (precision)
Real objects with a detection nearby	13 / 13	Real objects with any detection within 0.5 m
'Chair' labels / real chairs	113 / 1	Detections labelled chair / chairs in the scene
'Chair' labels on the real chair	1	Chair-labelled detections within 0.5 m of the real chair

6. Processing time and resources

Metric	Run 1	What it measures
Control-loop step (median / p95 / max)	91 / 380 / 2248 ms	Wall time of one whole control step (all agents)
Real-time factor	0.32	Simulated seconds per wall-clock second (1 = real time)
Wall-clock time	5666 s	Time the computer needed for the run
Process CPU (mean)	86 % of one core	CPU use of the Python process; simulator excluded
Memory (mean / peak)	1037 / 2973 MB	Resident memory of the Python process
GPU	not used	No part of the pipeline runs on the GPU

Agent (median / p95 / max, ms)	Run 1
PerceptionAgent	9.14 / 17.43 / 1986
SLAMAgent	10.62 / 23.64 / 85
MappingAgent	20.30 / 36.93 / 179
FrontierAgent	155.00 / 220.89 / 577
PlanningAgent	0.42 / 127.62 / 1061
LocomotionAgent	7.25 / 10.72 / 52
SupervisorAgent	0.02 / 0.03 / 0

Time of one call to each agent's step function.

7. Component ablation (offline replay of each recorded run)

The same recorded sensor data is replayed with one component switched off; the live robot is not changed. Values: ATE RMSE (cm) / heading error (deg).

Configuration	Run 1
full system	31.0 / 0.74
without scan matching (odometry + gyro only)	264.8 / 44.62
without loop closure	54.4 / 0.71
without degeneracy filter	32.9 / 0.74
without Manhattan heading prior	168.9 / 18.78

Comparison with published systems

Published values are quoted from the cited papers and product pages; 'derived' means two published numbers were divided. Test scenes, sensors, ground truth and map representations differ from ours as listed in the Test conditions column.

8. SLAM as a whole: pose accuracy, map accuracy and cost

System	Test conditions	Pose: ATE RMSE (cm)	Map: accuracy / completion (cm) / completion ratio (%)	Tracking per frame (ms)	Mapping per frame (s)	Map memory (MB)
Ours, Run 1	simulated 8-room arena, 158 m; forward RGB-D to 2D scan + gait odometry + gyro; 2D occupancy grid scored against real scene walls and objects; CPU [this work]	31.0	10.6 / 9.2 / 42.0	10.6	0.020	0.19
MNE-SLAM	Replica scenes, RGB-D, neural 3D map, GPU [1]	0.398	1.681 / 1.509 / 97.0	22.1	0.14	24.45
Loopy-SLAM	Replica scenes, RGB-D, neural 3D map, GPU [1]	0.369	1.593 / 2.645 / 91.1	-	-	-
GO-SLAM	Replica scenes, RGB-D, neural 3D map, GPU [1]	0.404	2.091 / 2.415 / 93.2	-	-	-
ESLAM	Replica scenes, RGB-D, neural 3D map, GPU [1]	0.657	2.082 / 1.754 / 96.0	68.5	0.29	20.73
Co-SLAM	Replica scenes, RGB-D, neural 3D map, GPU [1]	1.059	2.104 / 2.082 / 93.4	54.8	0.15	26.39
Point-SLAM	Replica scenes, RGB-D, neural 3D map, GPU [1]	0.559	1.935 / 2.779 / 90.5	29.7	10.47	31.42

9. Pose accuracy in multi-room scenes

System	Test conditions	ATE RMSE (cm)
Ours, Run 1	own 8-room arena, 158 m [this work]	31.0
MNE-SLAM	Replica multi-room apartments 1 / 2 / 0 [1]	1.02 / 0.59 / 0.48
ESLAM	Replica multi-room apartments 1 / 2 / 0 [1]	1.17 / 0.79 / 0.76
Loopy-SLAM	Replica multi-room apartments 1 / 2 / 0 [1]	1.43 / 0.60 / 0.64
ORB-SLAM3	Replica multi-room apartments 1 / 2 / 0 [1]	4.93 / 1.36 / 1.07
Swarm-SLAM	Replica multi-room apartments 1 / 2 / 0 [1]	5.56 / 5.61 / 1.80
CP-SLAM	Replica multi-room apartments 1 / 2 / 0 [1]	5.73 / 1.85 / 0.91

10. Pose accuracy on legged robots

System	Test conditions	Path (m)	ATE RMSE (cm)	ATE / path (%, derived)	CPU (%) / RAM (GB)
Ours, Run 1	simulation, simulator pose [this work]	158	31.0	0.20	86 / 1.0
A1 SLAM	real Unitree A1, lab, motion capture [3]	5.0-5.3	4.3-14.5	0.8-2.9	-
Cartographer	real Unitree A1, lab, motion capture [3]	5.0-5.3	4.0-139.7	0.8-27.9	-
FAST-LIVO2, LiDAR + camera + IMU (Honeywell)	real ANYmal D, mostly outdoor, 7 missions [4]	145-332	10.5	0.04	125 / 3.91
ORB-SLAM3, stereo (global shutter)	real ANYmal D, mostly outdoor, 7 missions [4]	145-332	33.0	0.14	290 / 1.16
RTAB-Map, stereo (global shutter)	real ANYmal D, mostly outdoor, 7 missions [4]	145-332	57.9	0.25	116 / 0.75

System	Test conditions	Path (m)	ATE RMSE (cm)	ATE / path (%, derived)	CPU (%) / RAM (GB)
RTAB-Map, RGB-D (ZED2i)	real ANYmal D, mostly outdoor, 7 missions [4]	145-332	241.3 (successful runs only)	1.03	-
ORB-SLAM3, RGB-D (ZED2i)	real ANYmal D, mostly outdoor, 7 missions [4]	145-332	821.4	3.50	-

11. Processing time per update

System	Quantity	Value	Hardware / note
Ours, Run 1	SLAM update, median	10.6 ms	CPU only, AMD Ryzen 7 7735HS with Radeon Graphics [this work]
ORB-SLAM2 (RGB-D)	tracking per frame	25.58 ms	Intel Core i7-4790, TUM fr3_office [2]
MNE-SLAM	tracking per frame	22.1 ms	authors' hardware [1]
A1 SLAM / Cartographer	optimization per scan	1.65 / 0.96 ms	optimizer call only [3]

12. Exploration speed

System	Test conditions	Distance (m)	Time (s)	Average speed (m/s)
Ours, Run 1	quadruped, forward RGB-D, 10 x 10 m arena [this work]	158	1800 (not completed)	0.088
DSVP	wheeled robot, 16-channel 3D LiDAR, Gazebo indoor, max 2 m/s [5]	1511	931	1.62 (derived)
TARE	wheeled robot, 16-channel 3D LiDAR, Gazebo indoor, max 2 m/s [5]	1209	658	1.84 (derived)
ARiADNE	wheeled robot, 16-channel 3D LiDAR, Gazebo indoor, max 2 m/s [5]	1053	610	1.73 (derived)
HEADER	wheeled robot, 16-channel 3D LiDAR, Gazebo indoor, max 2 m/s [5]	1030	576	1.79 (derived)
FARE (LLM + RL)	wheeled robot, 16-channel 3D LiDAR, Gazebo indoor, max 2 m/s [5]	1048	590	1.78 (derived)
ANYmal (product page)	quadruped, 360 deg LiDAR + depth cameras [7]	-	-	0.75 normal walking
Spot (product page)	quadruped, 360 deg terrain sensing [6]	-	-	1.6 maximum
Go2 (product page)	quadruped, 4D LiDAR [8]	-	-	about 5 maximum (lab)

13. Market systems: capabilities (from manufacturer pages; not measured by us)

System	Platform	Sensing	Explores unknown space	Needs prior map or path	Speed	Endurance	Compute	Protection
Ours, Run 1	quadruped (simulation)	forward RGB-D camera, gait odometry, gyro	Yes, frontier-based (did not complete in our test)	No	0.088 m/s measured	not measured (simulation)	CPU laptop, 86 % of one core	-
Boston Dynamics Spot [6]	quadruped	360 deg terrain sensing (4 m range); payload ports	No (needs a recorded map)	Yes (recorded map / route)	1.6 m/s max	~90 min (564 Wh)	not stated	IP54

System	Platform	Sensing	Explores unknown space	Needs prior map or path	Speed	Endurance	Compute	Protection
ANYbotics ANYmal [7]	quadruped	360 deg LiDAR, 6 depth cameras, 2 teleop cameras; thermal/zoom payload	Not stated on product page	Not stated	0.75 m/s normal	90-120 min (2 km)	2x Intel Core i7	IP67
Unitree Go2 (AIR/PRO/X) [8]	quadruped	4D LiDAR L2 (360 x 96 deg); depth camera on EDU only	No (user specifies the path)	Yes (user-specified path)	~5 m/s max (lab)	1-2 h (2-4 h EDU)	8-core CPU (EDU: optional Orin)	not stated
Unitree B2 [9]	quadruped	3D LiDAR x1, depth cameras x2, optical cameras x2 (varies by configuration)	Not stated	Not stated	> 6 m/s (special configuration; limited in practice)	4-6 h	Intel Core i5 / i7; optional Orin NX	IP67
DEEP Robotics X30 [10]	quadruped	integrated perception (sensors not listed)	Not stated	Not stated	>= 4 m/s	2.5-4 h	not stated	IP67
Emesent Hovermap (payload, mainly drones) [11]	sensor payload	LiDAR	Yes (autonomous exploration mode)	Not stated	not stated	not stated	onboard (not specified)	not stated

14. Like-for-like reference: TUM RGB-D Pioneer robot sequences, ATE RMSE (m)

Public dataset: Kinect on a wheeled Pioneer 3 robot looking forward, planar motion, motion-capture ground truth. Published results [12]; our system has not yet been run on these sequences.

System	pioneer_360	pioneer_slam	pioneer_slam2	pioneer_slam3	Average
Ours	not yet run	not yet run	not yet run	not yet run	not yet run
ORB-SLAM3 (RGB-D) [12]	1.059	2.118	1.709	1.992	1.720
OpenVSLAM (RGB-D) [12]	0.048	0.188	1.224	0.106	0.392
RTABMap (RGB-D) [12]	0.038	0.351	0.044	0.716	0.287

Map accuracy / completion / completion ratio: mean obstacle-to-obstacle distances and share within 5 cm. Ours are measured on 2D grid cells against the real scene geometry, published values on 3D reconstructed surfaces. ATE: absolute trajectory error (RMSE).

References

[1] T. Deng, G. Shen, C. Xun, S. Yuan, T. Jin, H. Shen, Y. Wang, J. Wang, H. Wang, D. Wang, W. Chen, "MNE-SLAM: Multi-Agent Neural SLAM for Mobile Robots", CVPR 2025, pp. 1485-1494. https://doi.org/10.1109/CVPR52734.2025.00146
[2] R. Mur-Artal, J. D. Tardos, "ORB-SLAM2: an Open-Source SLAM System for Monocular, Stereo and RGB-D Cameras". https://arxiv.org/abs/1610.06475
[3] J. Chen, F. Dellaert, "A1 SLAM: Quadruped SLAM using the A1's Onboard Sensors". https://arxiv.org/abs/2211.14432
[4] R. Corlito, F. Schmidt, N. Seibert, et al., "Sensor Configuration Matters: A Systematic Evaluation of Multimodal SLAM on Quadruped Robots" (ANYmal D, GrandTour dataset). https://arxiv.org/abs/2606.19067
[5] S. Liao, X. Lv, J. Lew, S. Zhang, et al., "FARE: Fast-Slow Agentic Robotic Exploration". https://arxiv.org/abs/2601.14681
[6] Boston Dynamics, Spot product specifications and Spot SDK 5.2.0 autonomy documentation (GraphNav, Directed Exploration). https://bostondynamics.com/products/spot/
[7] ANYbotics, ANYmal product page and technical specifications. https://www.anybotics.com/robotics/anymal/
[8] Unitree Robotics, Go2 product page and specification table. https://www.unitree.com/go2
[9] Unitree Robotics, B2 product page and specification table. https://www.unitree.com/b2
[10] DEEP Robotics, X30 product page and specification table. https://en.deeprobotics.cn/en/index/product3.html
[11] Emesent, Hovermap autonomy page. https://www.emesent.com/autonomy
[12] A. Merzlyakov, S. Macenski, "A Comparison of Modern General-Purpose Visual SLAM Approaches". https://arxiv.org/abs/2107.07589